

Final Project

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Dataset Introduction

The purpose of this project is to analyze HVAC operational data and identify opportunities to optimize RTU energy use. The data set used was provided by ASHRAE and contains room-level data from a terminal unit serving a 3-story building. The original datasets were recorded in 10-minute, 15-minute, and 30-minute intervals and were combined for analysis throughout this study.

The main objective of this project is to determine which variables contribute most toward RTU energy optimization. This paper covers the dataset-cleaning process, exploratory data analysis, machine learning models, optimization strategies, and the overall results discovered from the data.

Problem Statement

One of the main challenges in commercial HVAC systems is operating equipment efficiently while reducing energy usage. HVAC systems are among the largest contributors to commercial building electricity consumption, and while higher efficiency systems reduce long-term kW demand, they often increase upfront costs (Cho et al., 2018). Many RTUs operate inefficiently due to static scheduling, poor tuning, reactive maintenance, and limited real-time visibility into system performance. In many facilities, aging equipment and inconsistent monitoring make it difficult to detect inefficiencies before energy waste becomes significant.

Research Objectives

The objective of this project was to clean and analyze the dataset using exploratory data analysis and machine learning techniques to identify operational inefficiencies within the RTU

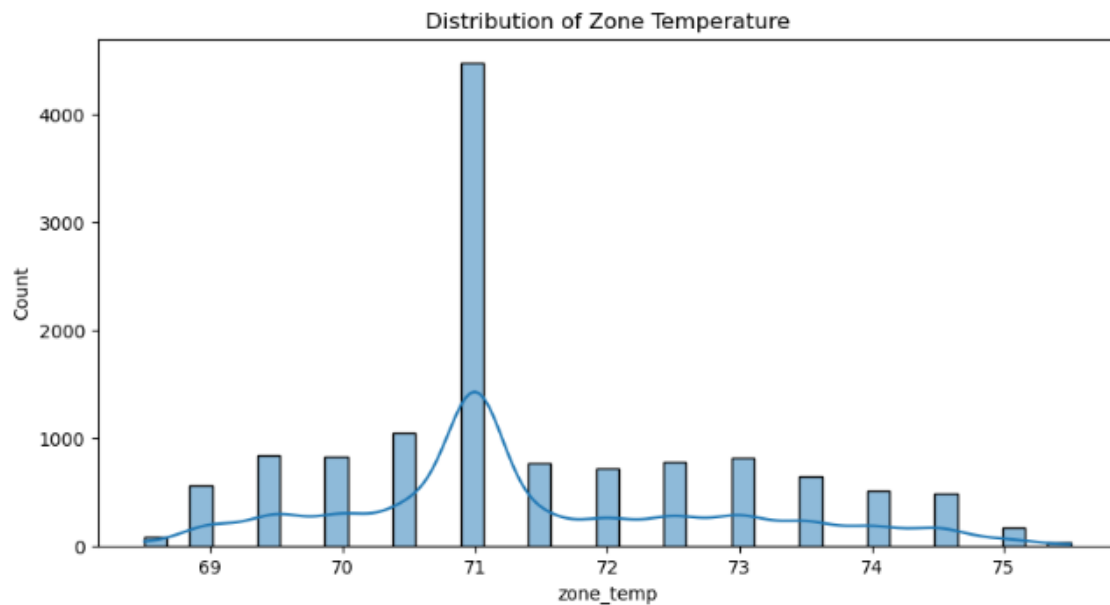
system. Statistical analysis and clustering methods were used to estimate real-time energy behavior and group operational patterns within the data.

Another goal was to apply machine learning models that can learn normal RTU behavior and identify deviations associated with inefficiencies. A comparison between the baseline model and advanced machine learning models was also performed to evaluate prediction accuracy and system analysis performance. Based on professional HVAC experience, historical operational data plays a major role in predictive analytics, setpoint optimization, and future upgrade decisions. The final objective was to recommend practical optimization strategies that move away from static scheduling and toward more adaptive and predictive HVAC control methods.

Exploratory Data Analysis

After combining the datasets, the first step was identifying missing values and inconsistent data points. HVAC datasets commonly contain missing information due to sensor failures, schedule changes, communication loss, or unreliable trends. The quality of the dataset was evaluated before proceeding with analysis to determine whether enough reliable data existed for modeling. A histogram was then created to analyze the distribution of zone temperatures and identify where temperatures were most heavily concentrated. The results showed the overall temperature tendency within space. Since the dataset only represented one room instead of an entire floor, the temperature averages and distribution may differ if additional zones were included in the analysis.

time	zone_temp	room_temp	OAT	DAT	SAT	zone_setpoint	setpoint_max	setpoint_min	clg_sp_current	htg_sp_current	clg_sp_offset	htg_sp_offset	occupied_c
2021-04-16 23:45:00	70.0	70.3	48	81.1	55.1	72.0	85	65	74.0	71.0	3	0	
2021-04-16 23:46:00	70.5	70.5	47	80.3	55.0	72.0	85	65	74.0	71.0	3	0	
2021-04-16 23:47:00	70.5	70.6	47	80.7	54.4	72.0	85	65	74.0	71.0	3	0	
2021-04-16 23:48:00	70.5	70.7	47	82.8	54.3	72.0	85	65	74.0	71.0	3	0	
2021-04-16 23:49:00	70.5	70.8	48	84.9	54.7	72.0	85	65	74.0	71.0	3	0	



Baseline Model vs Machine Learning Models

The baseline model was created first to provide a simple reference point before applying more advanced machine learning techniques. It used basic relationships within the HVAC data, mainly outdoor air temperature and overall system trends, to establish general system behavior. This helped determine whether the machine learning models were improving prediction accuracy and operational analysis.

Although the baseline model captured general HVAC behavior, it struggled to represent the more complex operating conditions found in real HVAC systems. RTU performance is influenced by multiple variables simultaneously, including outdoor air temperature, supply air temperature, zone temperature, occupancy schedules, and runtime conditions.

Several machine learning models were then used to better analyze RTU performance and identify inefficiencies. Linear Regression was used as the initial benchmark model to evaluate relationships between cooling load, outdoor air temperature, and setpoints. ARIMA and SARIMAX models were implemented because HVAC systems operate heavily around time-based patterns and occupancy changes throughout the day.

K-Means Clustering was used to group RTU behavior into categories such as efficient operation, inefficient operation, and transitional states. An LSTM neural network was also implemented because HVAC systems rely heavily on previous operating conditions, making LSTM models effective for learning long-term sequential behavior. Out of all models tested, the LSTM produced the strongest results due to its ability to track time-dependent HVAC behavior more accurately.

Optimization Strategies & Results

Based on the model results, several optimization opportunities were identified to improve RTU efficiency and reduce energy usage. Key areas included schedule improvements, setpoint adjustments, and monitoring how the units respond during changing load conditions throughout the day.

The LSTM model helped establish what normal system behavior looked like over time and identified periods where performance deviated from expected operation. Clustering analysis also

helped separate the RTU behavior into efficient operation, high-load conditions, and transitional states. Overall, the results showed that the system performs most of the time efficiently, while inefficiencies primarily occur during changing load conditions and periods of increased system stress.

Conclusion

This project demonstrated how machine learning and data analytics can be applied to real Building Management System data to better understand RTU performance and energy usage. While the baseline model provided a general understanding of HVAC behavior, the advanced machine learning models, especially the LSTM model, performed better because they were able to learn long-term and time-dependent system patterns.

In the future, I would expand my datasets, retrieve my own datasets, and expand the machine learning model a bit more. I would treat HVAC data the same as trading prices, since they operate similarly enough and are very similar in their use of historical data.

The results showed that predictive analytics can help identify inefficiencies, estimate cooling demand, and support improved HVAC optimization strategies. This project also demonstrated the future potential of predictive maintenance, real-time fault detection, and adaptive HVAC control strategies to reduce energy costs and improve overall system performance.

References

Cho, J., Lim, T., Kim, B. S., & Yun, G. Y. (2018). *Energy-cost analysis of HVAC system for office buildings*. *Energy and Buildings*, 173, 396–405.

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